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From Personality Traits to Personality Dynamics: New Approaches to Personality Research in Organizations

Creativity as a Function of Daydreaming and Cognitive Demands at Work: The Role of Openness to Experience and Neuroticism Personality States

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ABSTRACT

Recent research in organizational behavior has begun to focus on the role of daydreaming in the workplace, which refers to the spontaneous shift of attention from the external environment to internally generated thoughts. Emergent research suggests that daydreaming evolving from cognitive demands at work may serve as a precursor to creativity. However, despite this incipient interest, whether, how, and why different forms of cognitive demands lead to discrete forms of daydreaming and, therefore, creative outcomes in organizations remain in their early stages. Drawing on the Situation-Based Contingency of the Personality Manifestation Model, we propose that personality states, specifically openness to experience and neuroticism, have a pivotal role in explaining the effects of problem-solving and monitoring demands on problem-constructive daydreaming and guilt and fear-of-failure daydreaming, which, in turn, have varying effects on creative thinking. Our hypotheses were tested using data from a daily diary study conducted over one workweek with professional employees across diverse organizations. This study contributes to the emerging literature on daydreaming and personality states in organizations by illuminating their role in fostering creativity within the workplace.

Creativity is an advantage for most of today's organizations. New ideas, processes, and products increase organizational effectiveness and even survival. Thus, researchers in organizational behavior have dedicated efforts to identifying the factors that facilitate or hinder creative outcomes (George 2007). Building on this line of research, various individual differences, contextual characteristics, and psychological states have been found to be related to creativity in the workplace (Anderson et al. 2014). In this context, recent studies have begun paying attention to the role that daydreaming could play in employee creativity (Baer et al. 2021; Yu et al. 2024). As a component of the broader notion of mind-wandering, daydreaming refers to the process of drifting attention from the tasks at hand, such that individuals are involved in *off-task* thinking (McMillan et al. 2013; Smallwood

and Schooler 2015). Off-task thinking does not necessarily imply thinking unrelated to the task; instead, it involves momentary reflection and processing detached from the *immediate* execution of direct tasks (Dane 2018). In this regard, Baer et al.'s (2021) seminal studies indicated that daily cognitively demanding work can increase daydreaming activity, which, in turn, fosters creative thinking.

Nevertheless, as with any new field of scientific inquiry, this research stream still faces unexplored and unsolved issues. Cognitively demanding work is a multidimensional phenomenon involving different working conditions, such as problem-solving and monitoring demands (Wall et al. 1990). While problem-solving involves managing complexity and

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uncertainty, monitoring demands require sustained attention and concentration to prevent potential performance issues. Psychologically, these cognitive demands are associated with intensive information processing and attention management (Campbell 1988; Wall et al. 1990), though they have diverse psychological implications. As a type of challenge demand, problem-solving is often linked to motivation, whereas monitoring, as a potential hindrance, may be primarily associated with distress (cf. Lepine et al. 2005; Mazzola and Disselhorst 2019; Podsakoff et al. 2023). These demands may give rise to different forms of daydreaming.

Furthermore, most research on daydreaming and creativity has either overlooked the specific content of daydreams (e.g., Baird et al. 2012) or focused on particular types with positive consequences for creativity processes, such as problem-constructive daydreaming (Baer et al. 2021). However, foundational research on daydreaming suggests that daydreams can also involve content related to guilt and fear-of-failure concerning ongoing tasks (Huba et al. 1981; Li et al. 2022; Sun et al. 2022), which, we argue, could negatively impact workplace creativity. This type of daydreaming is linked to rumination and convergent thinking, both of which are potentially detrimental to creative ideation (Amabile et al. 2005; Vosburg 1998).

Additionally, there is a lack of knowledge regarding the psychological mechanisms that explain how cognitive demands trigger daydreams and their consequences for creativity. In task-oriented settings, the prevailing hypothesis is that daydreaming occurs when cognitive demands exceed the ability to stay focused, leading to task-unrelated thoughts or daydreams (McVay and Kane 2010). However, theoretical and empirical work on this topic remains underdeveloped in the context of organizational creativity. Specifically, more detailed descriptions and empirical testing are needed to understand the psychological processes linking cognitive demands to the occurrence of different types of daydreams in the workplace.

Addressing these issues is essential for advancing the theoretical understanding of the etiology and consequences of daydreaming in the workplace. Moreover, from a practical perspective, if different cognitive demands and types of daydreaming have distinct and potentially opposing relationships with creativity, organizations must recognize that not all aspects of cognitive demands and their outcomes are equivalent. Therefore, managing these factors requires specific intervention practices.

We propose that the relationship between daily cognitive demands, daydreaming, and creative thinking can be examined through the lens of personality states. These are contingent and dynamic expressions of personality traits over time, helping to explain how situational environmental conditions influence cognitive, affective, and behavioral processes within short time frames (Fleeson 2001). This perspective presents a more detailed exploration of these relationships. As an overarching framework, we draw on the Situation-Based Contingency of Personality Manifestation Model (Fleeson 2007), which posits that contingent contextual conditions contain active psychological elements that elicit corresponding personality states, leading to specific psychological processes and responses within the same time frame. In the case of this study, our adoption of this

theory leads to propose that daily problem-solving and monitoring demands are associated with openness to experience and neuroticism states, which in turn elicit problem-constructive daydreaming and guilt and fear-of-failure daydreaming and therefore increase or reduce creative thinking, respectively. As such, continuing with previous research on daydreaming in organizations (Baer et al. 2021), we make the case that personality states are pivotal psychological processes explaining why and how cognitive demands contribute to daydreaming and, consequently, creativity in organizations.

Overall, this study contributes to the organizational behavior literature in several ways. First, we expand the emerging research on daydreaming in organizations by investigating whether different cognitive demands lead to distinct forms of daydreaming activity. Second, we examine in detail whether specific dimensions of daydreaming activity are beneficial or detrimental to daily creative thinking. Finally, by addressing these questions, we introduce the construct of personality states, demonstrating their utility as psychological mechanisms that explain how cognitive demands influence daydreaming and creativity in the workplace. This approach also informs research on the dynamic perspective of personality, particularly regarding its application to organizational issues, such as understanding creativity at work. The following sections present our theoretical framework, the study conducted, and a discussion of the results.

1 | The Situation-Based Contingency of Personality Manifestation Model

Conventionally, personality is understood as a series of traits embedding stable psychological characteristics, over time and across situations, about how individuals think, feel, and behave (Hough et al. 2015). A long-lasting theoretical and empirical scientific tradition has proven the value of this construct in explaining motivational, attitudinal, and behavioral outcomes associated with job performance and well-being in the workplace (Barrick and Mount 1991; Judge et al. 2013). Notwithstanding, the notion of trait personality is also a matter of debate with the social cognitive approach highlighting that personality manifestation is situation dependent (Fleeson and Jayawickreme 2015; Mischel 2004; Mischel and Shoda 1995; Tett et al. 2021). This means that the expression of traits' psychological components is driven or inhibited by the context in which individuals perform. The Situation-Based Contingency of Personality Manifestation Model is a theoretical framework developed to account for the dynamic nature of personality expression (Fleeson 2007). This model is grounded in the concept of personality states, which are defined as momentary psychological qualities that reflect the elements of traits typically characterizing an individual. While personality traits describe the general consistent tendencies of a person, personality states are responses toward situational conditions that exhibit a significant degree of dynamism over minutes, hours, or days.

The situational approach makes significant assumptions that shape how personality is defined and understood. From the perspective of psychological states, personality dispositions are conceptualized as frequency distributions of states that share the same cognitive, affective, and behavioral characteristics

as traits, but not individuals' inner psychological properties that lead to these states (Fleeson 2001). These personality states emerge from situational drivers, and the consequences of these states are likely to be temporally bound to the period during which these states occur (Fleeson and Jolley 2006; Mischel and Shoda 1995). For example, within this framework, the trait of openness to experience is understood as an aggregation of states characterized by complex thinking, reflection, and curiosity (DeYoung et al. 2007). These states may emerge in response to daily task events that demand creative solutions, which, in turn, are likely to facilitate creative outcomes in daily life.

The first stage of the situation-based contingency theory addresses the etiology of personality states (Fleeson 2007). Psychological components activated by specific situations drive the emergence of personality states that align with the characteristics of those components. For instance, situations that require dedication, effort, or urgent solutions are associated with states of conscientiousness (Minbashian et al. 2010), whereas socially interactive environments give rise to states of extraversion and agreeableness (Huang and Ryan 2011). These examples illustrate that personality states involve a systematic relationship between the characteristics of a situation and a corresponding psychological process that promotes an adaptive response to situational demands (Allport 1937; Mischel 2004; Tett et al. 2021). Furthermore, this process suggests that the dynamic variability of personality states is a function of the fluctuations in contextual forces relative to their presence or absence over time.

The second stage of this theory concerns the consequences of personality states for other outcomes. Like stable personality traits, the cognitive, affective, and behavioral components of personality states drive specific psychological outcomes but are confined to the time frame in which these states occur. For example, conscientiousness states are expected to increase task proficiency (Barrick and Mount 1991; Minbashian et al. 2010), extraversion and agreeableness to improve interpersonal effectiveness (Chiaburu et al. 2011; Huang and Ryan 2011), and openness to facilitate creativity (Madrid et al. 2014; Zare and Flinchbaugh 2019). Conversely, state neuroticism can increase the likelihood of deviant behaviors (Bolton et al. 2010).

2 | Daily Problem-Solving Demands, Openness to Experience, Positive-Constructive Daydreaming and Creative Thinking

Cognitive demands are a central component of task characteristics, requiring active mental engagement and deep information processing. Cognitively demanding work encompasses various elements, including work constraints, task difficulty and challenge, memory and attention activities, uncertainty, and information load (Campbell 1988). Wall et al. (1990) offered a concise integration of these task characteristics, categorizing them into two primary factors, namely, problem-solving demands and monitoring demands. Originally, this model was designed to describe work in advanced manufacturing technology contexts. However, a detailed review of the model's factors suggests that problem-solving and monitoring requirements are applicable across a broader range of professional jobs. Problem-solving

demands involve navigating uncertainty in the task environment, managing difficult and complex problems, and addressing ambiguity and a lack of clarity regarding potential outcomes (Oldham and Cummings 1996; Perrow 1967; Wall et al. 1995; Zhou et al. 2012). On the other hand, monitoring demands involve intensive attentional processes that exert significant psychological pressure on employees. Specifically, monitoring demands pertain to tasks that require employees to maintain sustained focus and high levels of concentration to manage work constraints effectively and prevent incidents that could negatively impact performance (Wall et al. 1990, 1995). As such, in a broader sense, employees across diverse roles, particularly those involving knowledge management and development, are likely exposed to tasks with these characteristics.

While cognitive demands are often conceptualized as stable conditions under which employees perform their tasks (Jackson et al. 1993), alternative theoretical perspectives suggest that job demands are dynamic and subject to fluctuation, such as at the daily level (Daniels 2006; Downes et al. 2021). Drawing on this dynamic approach, we argue that the interplay between state openness to experience and daydreaming activity serves as a link between daily problem-solving demands and creativity. From a situation-based contingency perspective, daily problem-solving demands represent contextual conditions that characterize a workday involving tasks with greater uncertainty and ambiguity, as well as the need to manage complexity through open-ended problems that are not clearly defined. These active components of problem-solving align with intrapersonal elements of greater openness to experience, which is characterized by deeper intellectual engagement, open-minded thinking, curiosity, and approach motivation (Madrid et al. 2014; Tellegen and Atkinson 1974). In contrast, limited openness to experience is associated with a rigid cognitive organization centered on preferences for conventional solutions suited to well-structured problems (McCrae 1987; McCrae and Costa 1997). As such, heightened states of openness should be adaptive for addressing problem-solving demands by actively seeking a variety of solutions tailored to the task at hand.

In turn, states of openness are likely to be positively associated with daydreaming activity, particularly positive and constructive daydreams involving thoughts that are rich in cues for identifying potential pathways for managing tasks (Huba et al. 1981) and are characterized by cognitive flexibility and exploratory information processing (Dane 2018; Zhiyan and Singer 1997). Therefore, openness states are likely to foster positive-constructive daydreaming by intensifying inner mental activity and promoting reflection and cognitive elaboration on thought processes (McCrae and Costa 1997). Additionally, openness boosts approach motivation to tackle challenges at work and develop novel problem-solving strategies (McCrae and Costa 1997). Therefore, states of openness to experience would serve as a psychological mechanism that facilitates the positive relationship between problem-solving demands and positive-constructive daydreaming, as posited in the following hypothesis:

Hypothesis 1. *At the day-level, openness to experience states will mediate the positive relationship between problem-solving demands and positive-constructive daydreaming, such*

that problem-solving demands will be positively related to openness states, which in turn will be positively related to positive-constructive daydreaming.

As part of the psychological process outlined above, we propose that daily positive-constructive daydreams, arising from problem-solving demands and states of openness, will be positively associated with daily creative thinking. Creative thinking is a dynamic form of creativity involving the generation of unconventional ideas to solve problems, such as those encountered in the workplace (Amabile et al. 2005). Given its fluctuating nature, this construct is particularly suited for examining how daily demands, personality states, and daydreaming activities contribute to creativity on a daily basis. Supporting this idea, Baer et al. (2021) found that problem-solving demands are linked to creativity through daydreaming, emphasizing that solutions emerge not only from conscious effort but also during unexpected moments when individuals engage in other mental activities. Positive-constructive daydreams involve spontaneous thoughts that encompass content related to the problem at hand as well as material unrelated to it (Huba et al. 1981). This cognitive diversity supports incubation processes and the cross-fertilization of knowledge and ideas, ultimately fostering the development of original concepts (Baird et al. 2012; Zedelius and Schooler 2016).

Thus, positive-constructive daydreaming is positioned as an explanatory mechanism for the positive association of state openness and creativity:

Hypothesis 2. *At the day-level, positive-constructive daydreaming will mediate the positive relationship between openness states and creative thinking, such that openness states will be positively related to positive-constructive daydreaming, which in turn will be positively related to creative thinking.*

When the previous hypotheses are taken together, a two-stage serial mediation process is described in which the relationship between daily cognitive demands and creative thinking is explained by the experience of openness states and their concomitant daydreams. From the situation-based contingency framework, problem-solving demands contain active elements that foster an open mindset, expanding mental activity through positive-constructive thoughts, which by providing alternative task-related material, are positively associated with creative thinking.

Hypothesis 3. *At the day-level, openness states and positive-constructive daydreaming will mediate the positive relationship between problem-solving demands and creative thinking, such that problem-solving demands will be positively related to openness states, which in turn will be positively related to positive-constructive daydreaming, which in turn will be positively related to creative thinking.*

3 | Daily Monitoring Demands, Neuroticism, Guilt, and Fear-of-Failure and Creative Thinking

The study of monitoring demands in the context of daydreaming is particularly valuable because these demands inherently

concern the investment of attention in work tasks. As daydreaming involves the cognitive process of shifting and deploying attention away from immediate tasks (McMillan et al. 2013), it is expected to be sensitive to fluctuations in monitoring demands. Like problem-solving demands, monitoring demands are typically characterized as stable task characteristics features (Jackson et al. 1993). However, we propose that these demands can also exhibit daily fluctuations. For instance, over the course of a workweek, employees may encounter days where tasks require greater and more sustained attention, while on other days, the need for intensive monitoring may be less pronounced.

In our model, daily monitoring demands act as a situational contingency that leads to increased states of neuroticism, the activation of guilt and fear-of-failure daydreaming, and consequently, diminished creative thinking. Neuroticism is characterized by negative feelings, emotional instability, psychological volatility, and a tendency toward withdrawal (DeYoung et al. 2007). The emergence of neuroticism states from contingent monitoring demands is plausible due to the distressing nature of these pressures. Sustained attention can be psychologically taxing and may hinder performance, increasing stress and emotional strain (Teuchmann et al. 1999; Totterdell et al. 2006; Wall et al. 1990). As a result, psychological volatility and withdrawal are likely outcomes, often manifesting as negative emotionality and impulsivity (Judge et al. 2013).

A key tenet of the situation-based contingency of personality model is that personality states are adaptive responses to environmental situations (Fleeson 2007). While neuroticism is often associated with dysfunctional outcomes and reduced psychological well-being (Anglim et al. 2020), its core negative affective components challenge this view (Watson 2000; Watson and Clark 1992). From an affect-as-information perspective (Schwarz 2012), affect serves as signals about the environment, guiding adaptive responses. Negative affect, for instance, signals potential threats, prompting withdrawal behaviors and cognition to mitigate sustained negative consequences (Watson et al. 1999). Thus, negative affect can serve an adaptive function. Consequently, we argue that the affect-as-information theory should also apply to neuroticism states. This resolves the apparent tension regarding how a theory of personality states can explain states that, at first glance, are often regarded as non-adaptive. Applying this rationale, we propose that sustained concentration and attention on task performance can become psychologically threatening due to the risk of resource depletion. In this context, neuroticism serves as a signal of such risk, prompting responses that may protect the individual.

States of neuroticism, in turn, can increase the occurrence of daydreams characterized by guilt and fear-of-failure themes (Blouin-Hudon and Zelenski 2016; Huba et al. 1981). These daydreams often include thoughts of inefficacy in handling tasks and imagined scenarios of failure. Building on the preceding discussion, such daydreaming activity can be interpreted as an adaptive response to perceived psychological risks. Consequently, guilt and fear-of-failure thoughts, along with their associated self-appraisals, align with the unpleasant experiences inherent in the negative affect, volatility, and withdrawal components of neuroticism states. This leads us to propose the following indirect effect:

Hypothesis 4. *At the day-level, neuroticism states will mediate the positive relationship between monitoring demands and guilt and fear-of-failure daydreaming, such that monitoring demands will be positively related to neuroticism states, which in turn will be positively related to guilt and fear-of-failure daydreaming.*

The next step for the psychological process proposed refers to the mediation role of daily guilt and fear-of-failure daydreaming between state neuroticism and creative thinking. We anticipate that this type of daydreaming will have a negative impact on the development of unconventional ideas. The states of guilt and fear-of-failure are predominantly associated with rumination and convergent thinking—cognitive processes that are fundamentally opposed to the divergent thinking required for generating novel ideas (Amabile et al. 2005; Vosburg 1998). Research on affect suggests that negative emotions, such as those potentially present in daydreams fueled by guilt, fear, or worry, can also contribute to daily creative outcomes (Bledow et al. 2013; George and Zhou 2007). However, these effects are most evident in the early stages of the creative process, such as problem construction, rather than during the ideation phase. While negative affect may catalyze the recognition of problems and the need for creative solutions, it often hinders the subsequent generation of creative ideas (Baas et al. 2011).

Furthermore, from an adaptive perspective, guilt and fear-of-failure activity arising from monitoring demands and their associated neuroticism states may make individuals hesitant and inclined to avoid a creative approach. This is because working with novel ideas is often appraised as risky for task performance, as it involves concepts that have the potential to succeed but also entail a non-negligible risk of failure (Keith et al. 2024). Thus, considering the overall balance of these influences, we propose the following hypothesis:

Hypothesis 5. *At the day-level, guilt and fear-of-failure daydreaming will mediate the negative relationship between neuroticism states and creative thinking, such that neuroticism states will be positively related to guilt and fear-of-failure daydreaming, which in turn will be negatively related to creative thinking.*

The entire process described for daily monitoring demands constitutes a two-stage serial mediation, where the negative relationship between monitoring demands and creative thinking is mediated by states of neuroticism and guilt and fear-of-failure daydreaming. As a whole, from a situation-based contingency perspective, monitoring demands trigger neuroticism states as signals of threats related to possible psychological resource depletion. Consequently, heightened guilt and fear-of-failure daydreaming and reduced creative thinking can be understood as affective and cognitive responses to this potentially threatening scenario.

Hypothesis 6. *At the day-level, neuroticism states and guilt and fear-of-failure daydreaming will mediate the negative relationship between monitoring demands and creative thinking, such that monitoring demands will be positively related to neuroticism states, which in turn will be positively related to guilt and fear-of-failure daydreaming, which in turn will be negatively related to creative thinking.*

4 | Methods

4.1 | Procedure and Participants

We conducted a daily diary study to test our hypotheses. This research design was appropriate for investigating the model's constructs due to their tendency to fluctuate over time. Creative thinking, daydreaming, and personality states are dynamic processes that can vary from day to day (Amabile et al. 2005; Baer et al. 2021; Judge et al. 2014). Additionally, while traditionally conceptualized as stable components of job content, within-subject variation in job demands over short periods has also been demonstrated (Downes et al. 2021). Therefore, employing a daily diary study to capture this variability provides a more accurate representation of the constructs and allows for more precise statistical estimations compared to between-subject designs, which are better suited for studying trait-level constructs (Bolger and Laurenceau 2013; Ohly et al. 2010). Initially, we administered a survey to participants, obtaining their informed consent and collecting demographic data. The following week, from Monday to Friday, participants completed a daily survey at 4:00 p.m. This survey asked about their creative thinking, exposure to cognitive demands, states of openness to experience and neuroticism, and daydreaming activity throughout the day.

Participants were recruited in the LinkedIn professional social network by public posts inviting them to be part of the study in the researchers' profile. An initial pool of 123 individuals responded to some of the study's surveys. However, this number was reduced to 78 after excluding cases with incomplete data or participation in fewer than 2 days of the week-long panel. These 78 participants provided a total of 281 within-subject observations, with an average participation of 3.60 days (min. = 2, max. = 5). The gender of the participants was 50% male, their average age was 40.36 (SD = 8.83) years, and their level of education of them was 97% university studies. They were full-time employees of diverse organizations in the sectors of services (81.8%), consultancy (3.8%), and production (14.3%), with an average organizational tenure of 4.34 (SD = 4.36) years.

4.2 | Measures

4.2.1 | Problem-Solving and Monitoring Demands

Problem-solving demands were measured with three items of the Wall et al. (1995) scale adapted to measure the construct daily. The scale's stem was "today at work, to what extent have you ... ?," item example, "been required to deal with problems which were difficult to solve" (1: *not at all* to 5: *a great deal*, $\alpha = 0.82$). For monitoring demands, we adapted three items of the Wall's et al. (1996) scale, item example, "concentrated all the time to watch for things going wrong" ($\alpha = 0.73$).¹

4.2.2 | Openness to Experience and Neuroticism States

These personality constructs were measured with 12 items (six for each state) selected from the Goldberg's (1992) markers instrument to capture the core facets of these constructs according to the classification of DeYoung et al. (2007). The

scale stem was “think about today at work and rate to what extent you have been ...?” (1: *not at all* to 5: *to a great extent*), items examples, “reflexive” (intellect facet), and “curious” (openness facet) for openness ($\alpha=0.83$), together with “irritable” (volatility facet) and “insecure” (withdrawal facet) for neuroticism ($\alpha=0.86$).

4.2.3 | Daydreaming

This construct was measured with six items of the scale of Huba et al. (1981), three for each sub-dimension, adapted to measure daydreaming activity on a daily basis. Participants were asked, “today at work, to what extent have you ...?” (1: *not at all* to 5: *a great deal*), items example, “had a daydream that left me useful clues to situations I faced” (positive-constructive daydreaming, $\alpha=0.94$), and “imagined not being able to finish a job I am required to do” (guilt and fear-of-failure daydreaming, $\alpha=0.91$).

4.2.4 | Creative Thinking

This was measured with the three-item scale of Baer et al. (2021), which solicited participants to rate whether they think about related novel ideas in the workplace, with the stem “today at work, to what extent have you ...?” item example, “thought about some new and helpful ideas” (1: *not at all* to 5: *a great deal*, $\alpha=0.92$).

4.2.5 | Control Variables

To control for potential time serial dependence of endogenous variables across waves of data, the lagged factors from the previous day (−1 lag) for openness to experience, neuroticism, problem-constructive daydreaming, guilt, fear of failure, and creative thinking were included in the analyses. An additional advantage of this strategy was that the study

adopted a longitudinal design, as the statistical modeling was based on changes in the variables' scores across all study days. Furthermore, a linear time index variable was used as a control to account for any monotonic time trends over the study week, which could be associated with factors such as participant attrition, time-dependent covariates, and random fluctuations (Singer and Willett 2003). Thus, including the linear time index enhances statistical stability.

4.3 | Analytical Strategy

We followed a three-step strategy for data analysis. First, we conducted multilevel confirmatory analyses to determine the measures' validity (Byrne 2013). These analyses were conducted in a piecemeal fashion because the single model based on all measures underlying hypotheses testing demanded more parameters than the person-level observations. Accordingly, a series of within-subject factorial models were estimated to account for the different stages of the structural model proposed (Table 1) (Brown 2006). Models 1 and 2, and their comparison, tested if daydreaming facets and creativity were different constructs, given their apparent overlap. Model 3 examined the validity of personality states and daydreaming, while Model 4 analyzed the distinction between cognitive demands and personality states.

Second, we computed variance components from the null model, not estimating the hypotheses' relationships to determine the constructs' within-subject variability (Hox 2010). This is a necessary step in diary research to support conceptualizing variables as fluctuating over time. Intra-class correlation (ICC1) was used for these analyses, which informs the degree of between-subjects variance of measures. The within-subject variability of the constructs is described by the proportion computed based on $(1 - \text{ICC1}) * 100$.

Third, we tested the hypotheses using multilevel structural equation modeling with observed variables (path analysis) in

TABLE 1 | Confirmatory factor analysis models.

Model	N	χ^2	df	$\Delta\chi^2(\text{df})$	RMSEA	SRMR within	CFI	TLI
Model 1: Creative thinking, positive-constructive and guilt and fear-of-failure (three factors)	277/78	27.66	24		0.02	0.03	0.99	1.00
Model 2: Creative thinking, positive-constructive and guilt and fear-of-failure (one factor)	277/78	936.28	27**	908.62(3)**	0.35	0.22	0.40	0.21
Model 3: Positive-constructive, guilt and fear-of-failure, openness to experience and neuroticism [†]	269/78	289.74	125**		0.07	0.07	0.93	0.92
Model 4: Openness to experience, neuroticism, problem-solving demands, and monitoring demands ^{††}	272/78	254.67	125**		0.06	0.08	0.92	0.90

** $p < 0.01$.

[†]The reflexive, intellectual, and complex markers of openness to experience were allowed to freely covariate due to significant correlations between their errors.

^{††}The same strategy was implemented with moody, irritable, insecure, and shy markers of neuroticism. These issues were likely because the meaning overlaps between these markers within the same factors.

Mplus 8 (Muthén and Muthén 2019). We used Bayesian estimation due to its advantages in testing complex multivariate models, especially with small samples and unbalanced data, like the data utilized here (Kruschke et al. 2012). These analyses were based on noninformative prior distributions and 10000 iterations, allowing us to obtain 95% credibility intervals (CI) for the estimates in the model. The model utilized fixed intercept modeling, and all the predictor variables were person-mean centered to remove between-subject variance (Hox 2010), because our focus was on within-subject processes involving personality states. The model described two sequential mediation processes at the within-subject level. The first sequence included problem-solving demands, openness to experience states, positive-constructive daydreaming, and creative thinking. The second sequence comprised monitoring demands, neuroticism states, guilt and fear-of-failure daydreaming, and creative thinking. Indirect effects were also computed for all the mediation processes in the proposed full model based on the Monte Carlo method for assessing mediation with 95% CIs and 20000 repetitions (Preacher et al. 2010; Selig and Preacher 2008).

5 | Results

The results of confirmatory factor analyses supported the validity of the constructs tested in the model (Table 1), while variance components analysis showed that they fluctuated over days with within-subject variation values between 26% and 54% (Table 2). Means, standard deviations, zero-order correlations, and scale reliabilities are presented in Table 2.

Hypothesis 1 stated that openness to experience states would mediate the positive relationship between daily problem-solving demands and positive-constructive daydreaming, such that problem-solving would be positively related to openness, which in turn would be positively related to daydreaming. Results of within-subjects structural equation modeling presented in Table 3 and Figure 1 showed that problem-solving demands were positively related to openness, which in turn was positively related to positive-constructive daydreaming, with an indirect effect of 95% CI [0.053, 0.194]. Thus, Hypothesis 1 was supported. Hypothesis 2 held that constructive daydreaming

would mediate the positive relationship between daily openness states and creative thinking, such that openness would be positively related to daydreaming, which in turn would be positively related to creative thinking. The same model previously presented showed that in addition to the effect of openness on daydreaming, the latter was positively related to creative thinking, depicting an indirect effect of 95% CI [0.045, 0.245]. Hence, Hypothesis 2 was supported. Hypothesis 3 indicated that openness states and constructive daydreaming would mediate the relationship between daily problem-solving demands and creative thinking, such that problem-solving would be positively related to openness, which in turn would be positively related to daydreaming, which in turn would be positively related to creative thinking. Results indicated an indirect effect for this two-stage mediation of 95% CI [0.011, 0.076], which, therefore, supported Hypothesis 3.

Hypothesis 4 stated that neuroticism states would mediate the positive relationship between daily monitoring demands and guilt and fear-of-failure daydreaming, such that monitoring demands would be positively related to neuroticism, which in turn would be positively related to guilt and fear-of-failure daydreaming. Results showed that monitoring were positively related to neuroticism, which in turn was positively related to guilt and fear-of-failure daydreaming, with an indirect effect of 95% CI [0.085, 0.243]. Thus, Hypothesis 4 was supported. Hypothesis 5 held that guilt and fear-of-failure daydreaming would mediate the positive relationship between daily neuroticism states and creative thinking, such that neuroticism would be positively related to guilt and fear-of-failure daydreaming, which in turn would be negatively related to creative thinking. Results showed that guilt and fear-of-failure daydreaming were not related to creative thinking, such that the mediation proposed was not possible. Thus, Hypotheses 5 and 6 were not supported because the latter also depended on the same unsupported effect.

Finally, in a post-hoc analysis, a non-hypothesized mediation was observed in which monitoring demands were positively related to neuroticism states, which in turn were negatively associated with positive-constructive daydreaming, which in turn was related to creative thinking, with an indirect effect of 95% CI [−0.57, −0.002].

TABLE 2 | Means, standard deviations, intraclass correlations, zeroth-order correlations, and reliabilities.

Variables	<i>M</i>	<i>SD</i>	1-ICC1	1	2	3	4	5	6	7
1. Problem-solving demands	2.60	0.92	0.45	(0.82)	0.73**	0.30**	0.45**	0.34**	0.37**	0.17
2. Monitoring demands	3.28	0.77	0.44	0.44**	(0.73)	0.41**	0.46**	0.23*	0.31**	0.27*
3. Openness to experience states	3.08	0.71	0.43	0.40**	0.13*	(0.83)	0.08	0.39**	0.14	0.73**
4. Neuroticism states	1.89	0.76	0.26	0.31**	0.38**	0.00	(0.86)	0.19	0.71**	−0.12
5. Positive-constructive daydreaming	1.95	0.86	0.47	0.18**	0.04	0.47**	−0.05	(0.94)	0.41**	0.44**
6. Guilt and fear-of-failure daydreaming	1.97	0.99	0.29	0.21**	0.11	0.00	0.38**	0.14*	(0.91)	0.05
7. Creative thinking	3.12	0.81	0.54	0.35**	0.15*	0.47**	0.05	0.38**	0.04	(0.92)

Note: $N_{\text{within/between}} = 281/78$. Within-subject correlations are presented below the diagonal and the between-subjects correlations above the diagonal. Reliabilities are parenthesized and in bold in the diagonal.

* $p < 0.05$,

** $p < 0.01$.

TABLE 3 | Within-subjects Bayesian structural equation modeling for cognitive demands, personality states, daydreaming, and creative thinking.

Variable	Openness states			Neuroticism states			Positive-constructive daydreaming			Guilt and fear-of-failure daydreaming			Creative thinking		
	95% CI			95% CI			95% CI			95% CI			95% CI		
	EST	LL	UL	EST	LL	UL	EST	LL	UL	EST	LL	UL	EST	LL	UL
<i>Direct effects</i>															
Time index	−0.01	−0.07	0.05	0.01	−0.03	0.06	0.17	−0.02	0.14	−0.01	−0.06	0.06	0.01	−0.09	0.10
Lagged openness to experience ($t-1$)	−0.24	−0.38	−0.16												
Lagged neuroticism ($t-11$)				−0.01	−0.16	0.12									
Lagged positive-constructive daydreaming ($t-11$)							−0.21	−0.36	−0.09						
Lagged guilt and fear-of-failure daydreaming ($t-11$)										−0.25	−0.39	−0.12			
Lagged creative thinking ($t-11$)													−0.22	−0.37	−0.06
Problem-solving demands	0.28	0.15	0.38	0.02	−0.05	0.11	−0.04	−0.19	0.08	0.17	0.06	0.30	0.19	0.02	0.39
Monitoring demands	−0.12	−0.25	0.00	0.28	0.18	0.37	0.12	−0.03	0.31	−0.09	−0.22	0.02	0.03	−0.21	0.23
Openness to experience states							0.41	0.22	0.61	−0.10	−0.28	0.03	0.32	0.01	0.56
Neuroticism states							−0.27	−0.51	−0.02	0.56	0.33	0.75	0.03	−0.28	0.36
Positive-constructive daydreaming													0.30	0.13	0.49
Guilt and fear-of-failure daydreaming													−0.16	−0.37	0.08
Residual variance	0.14	0.11	0.18	0.09	0.07	0.11	0.23	0.20	0.29	0.16	0.14	0.20	0.36	0.28	0.46
Residual variance null model	0.23	0.19	0.26	0.13	0.11	0.16	0.37	0.32	0.42	0.24	0.20	0.28	0.58	0.47	0.70
R^2	0.22			0.13			0.36			0.23			0.56		

(Continues)

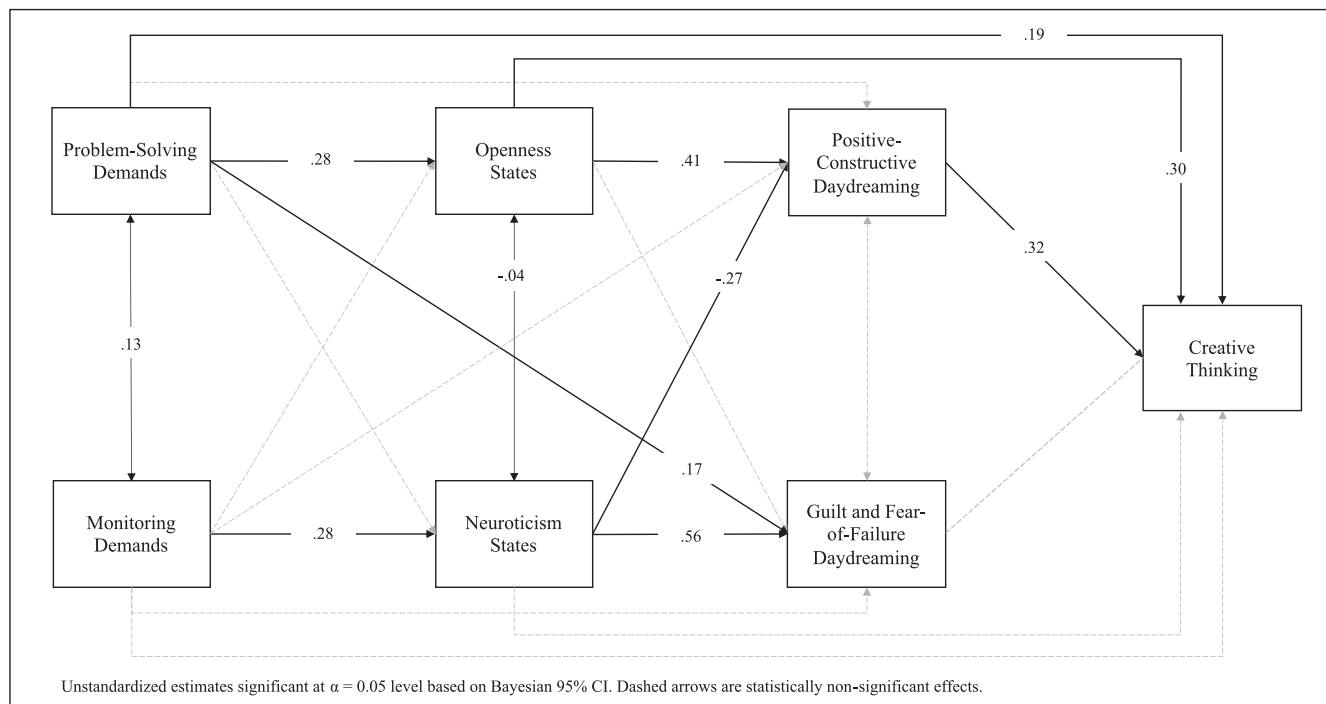


FIGURE 1 | Within-subjects Bayesian structural equation modeling for cognitive demands, personality states, daydreaming, and creative thinking.

6 | Discussion

The results of this study supported the idea that daily cognitive demands are associated with daydreaming and creativity in organizations, with personality states serving as explanatory mechanisms. Problem-solving demands positively predicted states of openness to experience, which facilitated positive-constructive daydreaming and, in turn, predicted creative thinking. In contrast, monitoring demands were positively linked to neuroticism states, which led to an increase in fear-of-failure daydreams; however, these daydreams were not related to creative thinking. Nonetheless, an exploratory analysis revealed that neuroticism negatively influenced positive-constructive daydreaming, which subsequently predicted creative thinking. Therefore, the mediation process originating from monitoring demands negatively affected creative thinking through heightened neuroticism states and reduced positive-constructive daydreaming.

Drawing on the Situation-Based Contingency of Personality Manifestation Model (Fleeson 2007), our model posits that daily job contingencies have active elements that can drive employees' personalities states influencing on processes of thinking involving daydreaming and creativity. Specifically, problem-solving and monitoring demands are likely linked to states of openness and neuroticism, respectively, due to their active components of uncertainty and psychological pressure (Wall et al. 1990). Once these states are activated, they initiate further psychological processes influenced by information processing and motivational mechanisms (Judge et al. 2014). Openness states are associated with a flexible mindset and an approach-oriented motivation to handle complex problems (DeYoung et al. 2007; McCrae and Sutin 2009), which explains the emergence of positive-constructive daydreams. In contrast, neuroticism states

are marked by emotional instability and a tendency toward withdrawal (Widiger 2009), which intensify guilt and fear-of-failure and reduced positive-constructive daydreams.

Subsequently, the positive-constructive daydreaming activity that arises from job demands and personality states can promote creative thinking. This form of daydreaming provides cues, knowledge, and potential pathways to deal with the problem at hand (Dane 2018; Huba et al. 1981), thereby fostering the ideation of novel solutions. In turn, creative thinking suffered when positive-constructive daydreaming was dampened by neuroticism states emerging from monitoring demands. However, the results concerning guilt and fear-of-failure daydreams present a challenge to our initial proposals. While their emergence from monitoring demands and neuroticism states aligns with our theoretical framework, this form of daydreaming activity did not reduce creativity as anticipated. We expected that the lack of broad, flexible, and divergent thinking typically associated with negative psychological experiences would be negatively associated with creativity. However, this statement alone does not fully explain the outcome, as the relationship between guilt and fear-of-failure daydreams and creativity was neither negative nor positive. Although these findings might suggest that no direct effects exist between these constructs, it is possible that moderators influence their occurrence. One potential intervening factor could be the stage of the creative process, particularly the distinction between creative problem construction and ideation (Reiter-Palmon and Robinson 2009). Evidence from studies on negative affect suggests that negative affect can enhance creativity during the problem construction phase (Bledow et al. 2013). Creative thinking often involves idea generation, but some degree of problem elaboration may still be present, which could obscure the potential impact of guilt and fear-of-failure daydreaming on creativity.

Alternatively, emotion regulation may act as a crucial moderator in the relationship between guilt and fear-of-failure and creative thinking. The consequences of the negative feelings associated with this style of daydreaming (e.g., guilt, fear, and worry) might depend on whether emotion regulation strategies are used to address the problem causing the negative affect or to directly alleviate the negative experience itself (Gross 1998). Creativity may be enhanced when individuals use problem-focused emotion regulation strategies, such as cognitive reappraisal (Lazarus and Folkman 1984). Conversely, if individuals employ emotion-focused strategies, such as suppression (Gross 1998), the negative experience may intensify, leading to emotional overwhelm and decreased performance, including reduced creativity.

Taking the above together, our results reveal that problem-solving demands have the potential to benefit creativity via openness states and positive-constructive daydreaming. In contrast, the cognitive demands associated with monitoring pressures are detrimental to creative thinking, as they give rise to neuroticism states that hinder positive-constructive daydreaming. As such, not all cognitive demands effects are created equally and have competing consequences when the focus is daydreaming and creative thinking.

From the task characteristics perspective, personality states reveal that diverse forms of cognitive demands emerge as significant antecedents of daydreaming activity in the workplace, encompassing their different forms described in classic literature on daydreaming (Huba et al. 1981; Singer 1975). In this context, daily problem-solving demands appear to foster daydreams that are functional for problem-solving, whereas monitoring demands may pose risks to the same outcomes. This distinction aligns with the differentiation between challenging and hindering job demands (Cavanaugh et al. 2000; Podsakoff et al. 2007). Challenges facilitate and promote the accomplishment of work tasks, whereas hindrances prevent, interfere with, or constrain task performance and the achievement of goals (Podsakoff et al. 2023). Accordingly, in our model, when employees encounter uncertainty and complexity during their workday, they are placed in contingent situations that engage their abilities and potentially transform them into psychological states and cognitions that support effective performance (challenge process). In contrast, the need for sustained attention can heighten emotional instability, ultimately posing risks to the effectiveness of work processes (hindrance process). Consequently, our findings extend existing knowledge on the implications of job demands for psychological functions in organizations. A long-lasting stream of research supports the influences of pressures at work on affective and cognitive distress and their consequences for work-related outcomes (Parker et al. 2017).

Here, we highlight that daily demands also contribute to off-task cognitive processes involving daydreaming. Within the domain of job and task characteristics, we adopted Wall et al.'s (1990) model, which focuses on cognitive pressures associated with task difficulty, challenge, uncertainty, and attention span. However, other cognitive pressures may also contribute to the emergence of personality states, daydreaming, and, consequently, workplace creativity. These include the need for memorization, flexibility, information processing, complex decision-making, and time-on-task.² For instance, time-on-task, characterized

by increasing rates of mind-wandering as task performance progresses (Zanesco et al. 2024), could interact with cognitive pressures to influence attentional dynamics and creativity outcomes. Future studies could explore these additional factors to expand understanding of the cognitive etiology underlying the psychological processes examined in our study.

This study also contributes to the field of research on mind-wandering in general, and daydreaming in particular, in organizational settings. Recent theoretical and empirical developments have confirmed that daydreaming frequently occurs during work (Ibaceta et al. 2024), consistent with findings that nearly 50% of our wakeful time is spent daydreaming (Killingsworth and Gilbert 2010). This prevalence suggests that daydreaming holds significant relevance for human adaptation, particularly in relation to creativity, as examined in our study. Our findings support that daydreaming encompasses varied content depending on the task environment, such that we show that not all forms of daydreams are equivalent in the organizational context (Dane 2018; Merlo et al. 2019). Distinctions between positive-constructive daydreams and guilt and fear-of-failure daydreams were established early in daydreaming research (Huba et al. 1981; Singer 1975). However, this differentiation has yet to be fully integrated into organizational research. Baer et al. (2021) examined different types of daydreams in the workplace, but their focus was limited to contrasting problem-oriented and bizarre daydreams as predictors of creativity. In contrast, our study not only replicates the value of daydreaming involving problem-solving but also highlights that guilt and fear-of-failure daydreaming is also a possible form of daydreaming at work. This distinction corroborates that positive-constructive daydreaming is the stronger predictor of creativity, both positively and negatively, depending on its driving cognitive demand. This contrasts with previous research that has broadly explored the benefits of daydreaming without considering its specific content (e.g., Baird et al. 2012). As Rodriguez-Boerwinkle et al. (2024) suggest, investigating the content of daydreams is crucial for fully understanding their relationship with creativity.

Another central contribution of this study is the adoption of daily personality states to explain psychological phenomena in the organizational realm. Traditionally, personality research in our field has focused on personality traits, which are considered stable psychological tendencies across time and situations (Hough et al. 2015). However, early psychological theorists like Allport (1937) and the seminal work of Walter Mischel over 50 years ago challenged this trait-centric approach (Mischel 1968), arguing that personality is highly situation-dependent. The core assumption of the socio-cognitive approach is that personality manifestation is not monolithic; rather, it is shaped by the contextual situation, which can either promote or inhibit certain personality expressions. Interest in these issues has grown recently, after a long period during which the question of how and why the environment shapes personality processes was overlooked (Fleeson and Jayawickreme 2015). The situation-based contingency model has embraced this line of inquiry, proposing that personality is not solely a matter of enduring traits but also includes dynamic states that evolve over time (Fleeson 2007). This theoretical perspective suggests that individuals are not always, for example, conscientious, extraverted, agreeable, open to experiences, or neurotic. Instead, an

accumulation of states tends to describe an individual's character at specific moments, indicating that personality characteristics may be more salient on some days than others. This approach could be helpful to advance research on personality and creativity showing that openness to experience has the most reliable connection to creative outcomes (Feist 1998; Hornberg and Reiter-Palmon 2017; Puryear et al. 2017; Tett et al. 2021). While task differences have been suggested as a possible explanation for some of these inconsistencies, our study indicates that conceptualizing personality in a dynamic fashion can expand this field of knowledge.

In methodological terms, we adopted a diary study design based on daily measures across a week to account for the dynamics of the constructs studied. As McCormick et al. (2020) pointed out, this kind of research has the advantage of examining the variability of the constructs over time, their lifespan and shape of effects, and whether the hypotheses tested are equivalent or differ when comparing the within and between-subject levels of analyses. Following this rationale, we corroborated that daily cognitive demands, personality states, daydreaming activity, and creative thinking are highly dynamic, with a fluctuation of around 50% over days. The exceptions were neuroticism states and guilt and fear-of-failure daydreaming, which showed limited variability during the same period. Consistently, the correlation between these variables was stronger at the between rather than the within-subject level of analysis. These results suggest greater complexity of the phenomenon studied. While the process associated with openness and positive-constructive daydreaming has clear dynamics, the effects regarding neuroticism and guilt and fear-of-failure may have more trait-like characteristics.

Furthermore, we conducted additional analyses to explore the lifespan of the hypothesized effects. Specifically, we estimated correlations between the variables and their scores observed on the previous day (−1 lag effects). This analysis was informative about whether the relationships between the variables occurred beyond the same day of their measurement and could have potential effects on the next day. Results indicated that, in most cases, these variables had no significant correlations (see Appendix S1). The exception for the relevant relationships hypothesized in the structural model were the modest association of openness states with previous-day problem-solving demands. Overall, these results, along with those supported by the structural equation model, provide initial evidence suggesting that the relationships between the examined variables occur within the same period due to the short lifespan of the constructs, which likely dissipate by the end of the day. This finding aligns with the principles of the situation-based contingency of personality manifestation theory (Fleeson 2007), which future research should re-examine to determine its robustness and generalizability. Moreover, negative carryover effects were observed over the course of the study week for problem-solving demands, openness to experience, problem-constructive daydreaming, and creative thinking. These effects are challenging to interpret, as they may reflect contextual conditions to which participants were exposed during the study, methodological issues such as inferences about the study's goals, habituation, contrast effects over time, or simply random variation. In substantive terms, these results underscore the importance of further research

on the constructs examined, emphasizing the need to consider short time spans to gain a more precise understanding of the dynamics between demands, personality states, daydreaming, and creative thinking. Theoretical developments and research designs should incorporate an event-based approach within working days to better capture the interplay between these variables. Insights from theories on episodic performance and experience sampling methodologies will be valuable in achieving these goals (Beal 2015; Beal et al. 2005; Dalal et al. 2014).

All of the above highlights the need to address how different or homologous the model tested at the within and between-subjects level of analysis is (McCormick et al. 2020). We did not have the chance to analyze a model with these characteristics because of the limited number of between-subject observations (participants). However, the low dynamics of some variables, the stronger between-subject correlations observed in some cases, and the short life span of effects over time lead to consider that future studies on this topic should consider two-level modeling. This approach will help account for the state-like and trait-like components of cognitive demands, personality variables, and daydreaming activity, offering a more comprehensive view of their interplay to explain creativity.

The implications of this research are not only conceptual but extend to practice. As previously mentioned, cognitive demands are often viewed as stable components of jobs, leading to static job design strategies (Parker and Wall 1998). However, recognizing the dynamic nature of these psychological pressures requires a more flexible approach to their management (McCormick et al. 2020). Insights from research and practice on affective events and states (Weiss and Cropanzano 1996), which, according to Fleeson and Jayawickreme (2015), are mirrored by personality states, provide relevant guidance. Leaders should adopt strategies delivering daily challenging but attainable problems, tasks, and goals as they can stimulate positive-constructive daydreaming, which is beneficial for creativity. In this context, leaders should be cautious about strictly penalizing employees for apparent distractions and counterproductive behavior (Bennett and Robinson 2000; Ibaceta et al. 2024; Merlo et al. 2019), as they may be in the process of incubating valuable solutions to organizational challenges. In contrast, engagement in extended periods of monitoring work should be carefully managed because its potential to produce neuroticism states rising distress processes and therefore reduced creative thinking.

6.1 | Limitations, Future Research, and Conclusion

As with any survey design, this study has limitations concerning the ability to infer causality between the examined variables. These limitations are primarily due to endogeneity issues, which relate to the possibility of omitted variables influencing the proposed model (Antonakis et al. 2010). For instance, variables such as creativity-related skills, task-related knowledge, working memory capacity, and metacognitive skills could act as confounding factors (Amabile 1988; Preiss 2022; Randall et al. 2014; Smeekens and Kane 2016). Endogeneity issues may also be involved in possible reverse causal effects, especially in the relationship between cognitive demands and personality states

(Hill et al. 2021). Theoretically, we followed the stressor-effect approach assumed in traditional and influential job content theories (Karasek 1979; Lazarus and Folkman 1984; Siegrist. 1996; Wall et al. 1990). Accordingly, job demands are expected to influence diverse affective processes and states, like information processing and motivation, leading to job-related outcomes (e.g., attitudes, behavior, and performance). However, accumulated research has also supported a strain-effect in the relationships among these variables, such that psychological states predict job demands perceptions (Ford et al. 2014; Guthrie et al. 2020). The perception hypothesis explains these effects by indicating that distressed employees (e.g., high in neuroticism states) can lead them to evaluate their jobs as more negative or challenging (De Lange et al. 2004). Similarly, openness to experience states may cause higher problem-solving demands perceptions but not the other way around, such that a more flexible mindset is associated with challenge appraisals.

To address the above issues, we conducted complementary analyses testing a model where personality states predicted job demands and, thereby, daydreaming and creative thinking. The results supported the effects of openness and neuroticism on problem-solving and monitoring demands, respectively. However, job demands were not associated with daydreaming activity, excepting a positive relationship between problem-solving and guilt and fear-of-failure daydreaming, suggesting that the causal direction from cognitive demands toward personality states offers a better explanation for the problem studied. Nevertheless, these analyses do not completely solve the endogeneity associated with possible reverse causality in the model. Thus, additional studies using instrumental variables and objective measures of cognitive demands are advisable to address these challenges for causal inference (Bollen et al. 2021; Hill et al. 2021).

Furthermore, we relied on self-reported ratings for all variables in our model. This approach was suitable for measuring openness to experience, neuroticism, daydreaming activity, and creative thinking, as these pertain to the inner psychological realm where employees are the primary witnesses of their own mental activity. Nevertheless, this method introduces common method variance issues (Podsakoff et al. 2012). Alternative methods can address these issues, such as objective cognitive demands and independent measures of creativity, like the supervisor ratings adopted in emergent research about daydreaming (Baer et al. 2021).

Finally, we focused on openness to experience and neuroticism states guided by previous research indicating that they are the primary personality variables for daydreaming (Ibaceta and Madrid 2021; Zhiyan and Singer 1997). We argued that openness and neuroticism should be the core personality states associated with daydreaming activity because these variables capture inner mental activity. Openness is a matter of deep intellect and imagination, while neuroticism involves emotional instability and withdrawal tendencies. However, from the Big-Five framework, state conscientiousness may be a relevant omitted variable in our model (Minbashian et al. 2010). Orderliness and achievement orientation embedded in this factor might reduce the prevalence of daydreaming by preventing distraction on the

current tasks. Thus, future research examining a broader set of personality states will extend the findings observed here.

In conclusion, this study aimed to provide a more comprehensive understanding of whether, how cognitive demands are associated with daydreaming and creativity via the processes offered by personality states. The results support this function, paving the way for new avenues of research and professional practice in organizational behavior and psychology.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. Online supporting information is available at https://osf.io/y5s6p/?view_only=17a86e720b814b92b2d011f121f8e06b.

Endnotes

¹ Measures adapted followed the criterion of using short scales in diary studies to not burden participants with a large instrument over the week of data collection and keep the engagement in the study. We use the Heggstad et al. (2019) recommendations to shorten the scales to ensure the content validity of the constructs studied.

² We thank the anonymous reviewer who highlighted these issues.

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